

No Bounded-Kernel RKHS at the Critical Holder Endpoint in Any Dimension

Agent lane 7

June 15, 2026

Source and Target

The source is Max Schöpple and Ingo Steinwart, *Which Spaces can be Embedded in Reproducing Kernel Hilbert Spaces?*, arXiv:2312.14711 [1]. In Section 4.2, Theorem 12 proves that for bounded subsets of $\mathbb{R}^d$ with non-empty interior there is no bounded-kernel RKHS

$$\text{Hol}^\alpha(X) \hookrightarrow H \hookrightarrow \ell_\infty(X)$$

when $\alpha < d/2$. The paper then says that this is tight up to the border case $\alpha = d/2$.

these preparations our result reads as follows.

Theorem 12. *Let (X, d) be a metric space and $0 < \beta \leq \alpha \leq 1$. If X is connected and there exists an RKHS H over X with*

$$\text{Höl}_d^\alpha(X) \hookrightarrow H \hookrightarrow \text{Höl}_d^\beta(X), \quad (12)$$

then there exists a constant $C > 0$ such that for all $0 < \delta < \min\{1, \text{diam } X\}$ we have

$$N_d^{\text{pack}}(X, \delta) \leq C\delta^{-2(\alpha-\beta)}. \quad (13)$$

Moreover, if there exists an RKHS H over X such that

$$\text{Höl}_d^\alpha(X) \hookrightarrow H \hookrightarrow \ell_\infty(X), \quad (14)$$

then there exists a constant $C > 0$, such that (13) holds for $\beta := 0$ and all $0 < \delta < \min\{1, \text{diam } X\}$.

Let us consider a bounded $X \subset \mathbb{R}^d$ with non-empty interior. By Theorem 12 and the packing number bound (8) we conclude that there is no RKHS H satisfying (12) if $\alpha - \beta < d/2$. Moreover, there is no RKHS H satisfying (14) if $\alpha < d/2$. Hence, (1) is infeasible for $d > 2$ within the family of Hölder spaces. We observe that the difference of the smoothness between the Hölder spaces in (12) needs to grow proportional to the underlying dimension in order to enable (1).

In Section 4.5, (26) will generalize Hölder spaces over X , allowing arbitrarily large non-integer smoothness. In Theorem 21 we will show that if $\alpha - \beta > d/2$, then there exists an intermediate RKHS H such that we have

$$\text{Höl}^\alpha(X) \hookrightarrow H \hookrightarrow \text{Höl}^\beta(X)$$

and in Theorem 22 we will show that for $\alpha > d/2$ there exists an intermediate RKHS H such that

$$\text{Höl}^\alpha(X) \hookrightarrow H \hookrightarrow C^0(X)$$

holds. Hence, the bounds derived in Theorem 12 are tight up to the border cases $\alpha - \beta = d/2$, respectively $\alpha = d/2$. For general metric spaces (X, d) however, we do not know any condition that implies the existence of an RKHS H satisfying (12) or (14).

The requirement that X is a *bounded* subset of $\mathbb{R}^d$ is crucial to obtain positive answers to (1), as the following

This packet settles the bounded-kernel critical endpoint $\alpha = d/2$ on cubes in every dimension, under the standard Holder endpoint conventions explained below.

Endpoint Convention

For non-integer $s > 0$, $\text{Hol}^s([0, 1]^d)$ denotes the usual bounded $C^{\lfloor s \rfloor, s - \lfloor s \rfloor}$ Holder class. For $s = 1$, it denotes the bounded Lipschitz class. At higher integer s , the source paper's generalized Holder notation is convention-dependent; the proof below applies to the classical C^s endpoint, to $C^{s-1,1}$, and to the Holder-Zygmund/Besov endpoint $B_{\infty, \infty}^s$. The common property used by the proof is the elementary character estimate

$$\|e_m\|_{\text{Hol}^s([0, 1]^d)} \leq C_{d,s}(1 + |m|)^s, \quad e_m(x) = \exp(2\pi i m \cdot x), \quad m \in \mathbb{Z}^d.$$

For $C^{k, \theta}$ this follows by differentiating e_m up to order $k = \lfloor s \rfloor$ and taking the θ -Holder seminorm of those derivatives; for Lipschitz and C^k it is immediate.

Result

Theorem 1. *Let $d \geq 1$ and put $s = d/2$. There is no bounded-kernel reproducing kernel Hilbert space H on $[0, 1]^d$ such that*

$$\text{Hol}^s([0, 1]^d) \hookrightarrow H \hookrightarrow \ell_\infty([0, 1]^d)$$

continuously.

In particular, the bounded-kernel Holder endpoint left open by the source paper has a negative answer for every dimension d .

Idea of the Proof

The proof uses only Fourier characters. If such an RKHS sandwich existed, then each periodic character $e_m(x) = \exp(2\pi i m \cdot x)$ would have a Hilbert feature vector Je_m . Bounded point evaluations give uniformly bounded Riesz representatives v_t , indexed by $t \in \mathbb{T}^d$.

For a finite frequency set $F \subset \mathbb{Z}^d \setminus \{0\}$, project the representatives v_t onto the finite-dimensional space spanned by $\{Je_m : m \in F\}$. The Fourier coefficients u_m of this projected feature map satisfy a Bessel bound

$$\sum_{m \in F} \|u_m\|^2 \leq C.$$

On the other hand, reproducing the character e_m , together with $\|e_m\|_{\text{Hol}^{d/2}} \lesssim |m|^{d/2}$, forces

$$\|u_m\| \gtrsim |m|^{-d/2}.$$

Thus the Bessel bound would imply

$$\sum_{m \in F} \frac{1}{(1 + |m|)^d} \leq C.$$

Letting F exhaust $\mathbb{Z}^d \setminus \{0\}$ contradicts the critical divergence of $\sum_{m \neq 0} |m|^{-d}$.

Finite Character Lemma

Let $\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$, with normalized Haar measure. We write the proof over complex scalars. A real factorization may be complexified in the usual way, so proving the complex statement rules out both real and complex RKHS sandwiches. Throughout this section Hilbert inner products are linear in the first variable.

Lemma 1. *Let E be a Banach space of functions on $[0, 1]^d$. Assume that for some $s > 0$ and $M > 0$, every character*

$$e_m(x) = \exp(2\pi i m \cdot x), \quad m \in \mathbb{Z}^d,$$

belongs to E and satisfies

$$\|e_m\|_E \leq M(1 + |m|)^s.$$

Suppose there are a Hilbert space $\mathcal{H}$ and bounded linear maps

$$J : E \rightarrow \mathcal{H}, \quad R : \mathcal{H} \rightarrow \ell_\infty([0, 1]^d)$$

such that $RJf = f$ for every $f \in E$. Put

$$a = \|J\|, \quad b = \|R\|.$$

Then for every finite $F \subset \mathbb{Z}^d \setminus \{0\}$,

$$\sum_{m \in F} \frac{1}{(1 + |m|)^{2s}} \leq a^2 b^2 M^2.$$

Proof. Choose the standard half-open fundamental-domain section $\sigma : \mathbb{T}^d \rightarrow [0, 1]^d$, and define

$$(Bh)(t) = Rh(\sigma(t)), \quad t \in \mathbb{T}^d.$$

Then $\|B\| \leq b$, and

$$BJe_m(t) = e_m(\sigma(t)) = e_m(t).$$

For each $t \in \mathbb{T}^d$, the functional $h \mapsto (Bh)(t)$ on $\mathcal{H}$ has norm at most b . Let $v_t \in \mathcal{H}$ be its Riesz representative, so that

$$(Bh)(t) = \langle h, v_t \rangle_{\mathcal{H}}, \quad \|v_t\|_{\mathcal{H}} \leq b.$$

Fix a finite set $F \subset \mathbb{Z}^d \setminus \{0\}$, and let

$$E_F = \text{span}\{Je_m : m \in F\} \subset \mathcal{H}.$$

Let P_F be the orthogonal projection onto E_F . The map

$$t \longmapsto P_F v_t$$

is a continuous E_F -valued function, because E_F is finite-dimensional and, for every $h \in E_F$, the scalar function $\langle h, P_F v_t \rangle = \langle h, v_t \rangle = (Bh)(t)$ is a finite linear combination of continuous characters.

For $m \in F$, define the vector-valued Fourier coefficient at index $-m$

$$u_m = \int_{\mathbb{T}^d} e_m(t) P_F v_t dt \in E_F.$$

Parseval's inequality for the continuous E_F -valued function $t \mapsto P_F v_t$ gives

$$\sum_{m \in F} \|u_m\|_{\mathcal{H}}^2 \leq \int_{\mathbb{T}^d} \|P_F v_t\|_{\mathcal{H}}^2 dt \leq b^2.$$

For each $m \in F$, since $Je_m \in E_F$,

$$1 = \int_{\mathbb{T}^d} e_m(t) \overline{e_m(t)} dt = \int_{\mathbb{T}^d} \langle Je_m, P_F v_t \rangle \overline{e_m(t)} dt = \langle Je_m, u_m \rangle.$$

Consequently,

$$1 \leq \|Je_m\|_{\mathcal{H}} \|u_m\|_{\mathcal{H}} \leq aM(1 + |m|)^s \|u_m\|_{\mathcal{H}}.$$

Thus

$$\|u_m\|_{\mathcal{H}}^2 \geq \frac{1}{a^2 M^2 (1 + |m|)^{2s}}.$$

Combining this lower bound with the Bessel upper bound proves the lemma. $\square$

Proof of Theorem 1

Assume, toward a contradiction, that such an RKHS H exists, and let

$$J : \text{Hol}^s([0, 1]^d) \rightarrow H, \quad R : H \rightarrow \ell_\infty([0, 1]^d)$$

be the two continuous inclusions, where $s = d/2$. After complexifying H, J, R if necessary, use the same notation for the corresponding maps on complex-valued functions.

The endpoint Holder norm satisfies the character estimate

$$\|e_m\|_{\text{Hol}^s([0, 1]^d)} \leq M(1 + |m|)^s$$

for a constant M depending only on d and the chosen equivalent endpoint norm. Therefore Lemma 1 gives

$$\sum_{m \in F} \frac{1}{(1 + |m|)^d} \leq \|J\|^2 \|R\|^2 M^2$$

for every finite $F \subset \mathbb{Z}^d \setminus \{0\}$.

Taking $F_N = \{m \in \mathbb{Z}^d \setminus \{0\} : |m| \leq N\}$ gives a uniform bound on the partial sums of

$$\sum_{m \in \mathbb{Z}^d \setminus \{0\}} \frac{1}{(1 + |m|)^d}.$$

But this series diverges. Indeed, the annulus $\{m \in \mathbb{Z}^d : 2^j \leq |m| < 2^{j+1}\}$ contains at least $c_d 2^{jd}$ lattice points for all large j , and each term on that annulus is at least $C_d 2^{-jd}$. Hence each dyadic annulus contributes a positive amount bounded below independently of j . This contradiction proves the theorem. $\square$

Verification and Novelty Notes

The proof is deterministic and uses no numerical computation. The only analytic inputs are the Riesz representation theorem in Hilbert space, finite-dimensional Hilbert projections, vector-valued Parseval/Bessel inequality on the torus [2], the elementary Holder norm estimate for Fourier characters, and the critical lattice-sum divergence $\sum_{m \in \mathbb{Z}^d \setminus \{0\}} |m|^{-d} = \infty$.

The local run indexes were searched for arXiv id 2312.14711 and for the endpoint keywords Hol to smoothness $d/2$, $\alpha = d/2$, bounded kernel, Holder-Zygmund, Besov critical endpoint, and Hilbert factorization. The exact hits were the previous two-dimensional packet, its bundled one-dimensional companion packet, and the superseded attempt note.

Bounded web searches for the source title plus **border**, for **bounded-kernel RKHS alpha=d/2 Holder**, for **Holder-Zygmund d/2 RKHS**, and for **Besov critical RKHS bounded kernel** found the source arXiv record and no exact later answer to this endpoint.

Scope and Remaining Border Cases

This packet settles the bounded-kernel endpoint

$$\text{Hol}^{d/2}([0, 1]^d) \hookrightarrow H \hookrightarrow \ell_\infty([0, 1]^d)$$

for every $d \geq 1$, in the standard endpoint conventions above. It does not settle every Holder-to-Holder border case

$$\text{Hol}^\alpha \hookrightarrow H \hookrightarrow \text{Hol}^\beta, \quad \alpha - \beta = d/2.$$

Those are harder classification questions and may have mixed positive and negative outcomes.

Human Review Recommendation

Review should focus on Lemma 1. In particular, check the finite-dimensional projection $P_F v_t$, the Fourier coefficient convention, and the reduction from a cube RKHS sandwich to bounded feature vectors for characters on the torus. Once these are accepted, the all-dimensional contradiction is exactly the critical divergence of $\sum_{m \neq 0} |m|^{-d}$.

References

- [1] Max Schöpple and Ingo Steinwart. *Which Spaces can be Embedded in Reproducing Kernel Hilbert Spaces?* arXiv:2312.14711, 2023. <https://arxiv.org/abs/2312.14711>.
- [2] Walter Rudin. *Fourier Analysis on Groups*. Wiley, 1962.